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Radar high-resolution range profiles target recognition based on stable dictionary learning

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Abstract: Sparse representation models based on dictionary learning have led to interesting results in signal restoration and target recognition. However, due to the redundancy defined by overcomplete dictionary atoms in new space, finding sparse representations from inaccurate measurements may cause uncertainty and ambiguity. Especially for radar automatic target recognition using high-resolution range profiles (HRRP), the target-aspect sensitivity, amplitude fluctuation and outliers in HRRPs could result in mismatch among the sparse representations of the same class and thus deteriorate the recognition performance. This article proposes a novel stable dictionary learning method to deal with this problem and improve the pattern recognition performance. The proposed method relies on the constraints that the sparse representations of adjacent HRRPs without scatterers' motion through range cells should have the same support and lower variance. The structured sparse regularisation is then used to automatically select the optimal dictionary basis vectors for stable sparse coding. Experiments based on the measured HRRP dataset validate the performance of the proposed method. Moreover, encouraging results are reported with small training data size and under different signal-to-noise ratio conditions.

1 Introduction

A high-resolution range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which contains abundant target structure signatures, such as target size, scatterer distribution, etc. Therefore, radar HRRP target recognition receives intensive attention from the radar automatic target recognition (RATR) community [1–13].

Feature learning is a key technique for RATR. A good feature can reduce the noise and redundancy in the original signal, and also be beneficial to extract data information of interest. Many researchers [6–11] explore various feature (representation) learning methods for HRRP-based RATR. Researchers [6–8] investigate the recognition methods based on spectra features, [9] further compares the recognition performance of high-order spectra features. Such methods process the recognition task in some representation spaces with specific physical meaning. In [10], PCA-based representation subspace is constructed to minimise the reconstruction error for RATR. Du *et al.* [11, 12] utilise factor analysis (FA) model to characterise and recognise HRRP in a low-dimensional latent representation space. These methods based on mathematical models design the classifiers using low-dimensional representations of signals, and usually perform well in practice. Nevertheless, all these methods have an underlying assumption: the representation dimension is not more than the maximal observation dimension. Recently, many researchers [13–23] suggest, however, that dictionary learning can achieve superior performance via using sparse overcomplete representation of signal in various real world problems.

Using an overcomplete dictionary adapted to the data, we can represent signals by sparse linear combinations of the dictionary atoms. Several interesting arguments, both heuristic and theoretical, have been advanced to support the benefits of dictionary learning. Researchers [13–23] show that dictionary learning is an efficient approach to address specific tasks such as natural image processing, data compression, etc. There are many advantages of working in this way, including the fact that it encourages a simple sparse model, and therefore overfitting is avoided. The overcomplete atoms are better able to capture data structures and patterns inherent, and the inferred sparse coefficients often have biological and physical meaning, which is of interest for model interpretation.

For dictionary learning methods, however, the problem of finding sparse representations from inaccurate measurements may cause uncertainty and ambiguity, because there is redundancy to some extent in new space defined by overcomplete dictionary atoms. That is a real issue for HRRP-based RATR. Due to the existence of target-aspect sensitivity, amplitude fluctuation and outliers in HRRPs, it is not easy to calculate stable sparse representations of inaccurate HRRPs via conventional dictionary learning methods. This instability leads to the mismatch among the sparse representations of the same class, thereby deteriorating the recognition performance. To deal with the issue, this paper focuses on a stable dictionary learning (SDL) method for radar HRRP target recognition task.

The stability conditions of sparse overcomplete representation with respect to a fixed dictionary are described at first, although the conditions would be unduly restrictive for HRRP-based RATR. Then, we develop a SDL method, via utilising similar but relaxed strategy, to learn an adaptive dictionary. The proposed method employs a robust objective function, which handles the stability by enforcing the sparse representations of adjacent HRRPs should have the same support and lower variance. The structured sparse regularisation is then used to automatically select the optimal dictionary basis vectors for stable sparse representation. The constraints stem from the structural similarity among adjacent HRRPs and are adopted to improve the ability of finding underlying sparse representations, which may be useful for learning a dictionary well-matched to the structure of signal and obtaining robust recognition performance. Another contribution of this paper is that efficient algorithms based on greedy strategy are proposed for the implementation of SDL. Experimental comparisons with three traditional HRRP-based RATR methods: adaptive Gaussian classifier (AGC) [3], PCA-based minimum reconstruction error approximation [10], FA model [11], and two dictionary learning methods: K-SVD [13], Method of optimal directions (MOD) [14],

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demonstrate that the proposed method has a superior recognition performance on the measured radar HRRP dataset. Furthermore, SDL's mechanism makes it possible to appropriately share the information among samples from different target-aspects and learn the global dictionary collectively, thus offering the potential to improve the overall recognition and generalisation performance even with small training data size.

The remainder of the paper is organised as follows. We briefly review the conception of dictionary learning in Section 2. The SDL method is proposed in Section 3. In Section 4, we discuss the training and classification algorithms for SDL in detail. The experimental results based on synthetic data and measured HRRP data are provided in Section 5, followed by conclusions in Section 6.

2 Review of dictionary learning

Dictionary learning is an interesting and promising research field. It focuses on the development of algorithms for building dictionaries that provide efficient sparse representations of signals.

2.1 Dictionary learning

Let $\boldsymbol{X}$ be a set of P-dimensional N inputs signals, that is, $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{P\times N}$. Learning an adaptive dictionary $\boldsymbol{D}$ with K items (implying $P < K$ to make $\boldsymbol{D}$ overcomplete) for sparse representation of $\boldsymbol{X}$ can be calculated by

$$\{\boldsymbol{D}, \boldsymbol{A}\} = \arg\min_{\boldsymbol{D},\boldsymbol{A}} \|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{A}\|_{\mathrm{F}}^2 \quad \text{s.t.} \quad \forall i, \quad \|\boldsymbol{\alpha}_i\|_0 \le S \tag{1}$$

where $\|\boldsymbol{\alpha}\|_0$ represents the l_0 norm, counting the number of non-zero elements in vector $\boldsymbol{\alpha}$, $\boldsymbol{A} = [\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_N] \in \mathbb{R}^{K\times N}$ is the sparse representation matrix of $\boldsymbol{X}$.

Several methods [16–23] have been developed to solve (1), all of which are the iterative algorithms to alternately optimise the following two stages.

2.1.1 Sparse coding: Given dictionary $\boldsymbol{D} \in \mathbb{R}^{P\times K}$ with l_2-normed columns, sparse coding computes the sparse representation $\boldsymbol{\alpha} \in \mathbb{R}^{K\times 1}$ of $\boldsymbol{x} \in \mathbb{R}^{P\times 1}$ by solving

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \quad \text{s.t.} \quad \|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{\alpha}\|_2^2 \le \varepsilon \tag{2}$$

or

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha}} \|\boldsymbol{x} - \boldsymbol{D}\boldsymbol{\alpha}\|_2^2 \quad \text{s.t.} \quad \|\boldsymbol{\alpha}\|_0 \le S \tag{3}$$

where ε and S represent the bounded representation error and sparsity level, respectively. Unfortunately, the exact solution of (2) or (3) requires searching over all combinations of non-zero elements, which is an NP-hard problem and hence computationally intractable.

Two approaches can be employed for approximating the solution of sparse coding. One approach is using greedy algorithm (GA), such as matching pursuit [24], orthogonal matching pursuit (OMP) [25]. Another is to relax the l_0 constraint to an l_1 constraint, yielding a convex quadratic programming problem, which is more precise but lower efficiency and can be solved by basis pursuit [26] or LARS-Lasso [27], etc. In several applications [13, 16, 17, 22, 23], GA provides sparser solutions and higher computational efficiency than convex relax methods.

2.1.2 Dictionary update: Compared with prespecified dictionaries, using dictionaries adapted to the data may better match the signal of interest, and lead to better performance for many practical tasks [14].

$$\boldsymbol{D} = \arg\min_{\boldsymbol{D}} \|\boldsymbol{X} - \boldsymbol{D}\boldsymbol{A}\|_{\mathrm{F}}^2 \quad \text{s.t.} \quad \forall k, \quad \|\boldsymbol{d}_k\|_2^2 = 1 \tag{4}$$

This constrained optimisation problem can be addressed using several methods, such as K-SVD, MOD and gradient descent with iterative projection.

2.2 Stable recovery of sparse overcomplete representations

With a learned dictionary, we can generate the sparse representations of signals, which provide more flexibility in signal representation and more effectiveness at various tasks. However, due to the redundancy of the dictionary, the problem of finding sparse representation is undetermined, thus the sparse representations may be unstable even with slightly inaccurate measurements. That can be a real issue for recognition task, in which our representation procedure must be stable: small changes in the observations should result in small changes in the sparse representations.

Several researchers [28–30] study this issue carefully. Considering a linear model $\boldsymbol{x} = \boldsymbol{x}_0 + \boldsymbol{n} = \boldsymbol{D}\boldsymbol{\alpha}_0 + \boldsymbol{n}$, where $\boldsymbol{x}_0 \in \mathbb{R}^{P\times 1}$ is the ideal noiseless signal and $\|\boldsymbol{n}\|_2 < \varepsilon$. Donoho *et al.* [28] have proved that when sufficient sparsity is present, that is, $\|\boldsymbol{\alpha}_0\|_0 < (1 + M^{-1})/2$, solving (2) enables stable recovery of $\boldsymbol{x}_0$ from contaminated observation $\boldsymbol{x}$. M is the mutual coherence of the dictionary $\boldsymbol{D}$.

$$M = M(\boldsymbol{D}) = \max_{1\le k,j\le K, k\ne j} |\boldsymbol{G}_{k,j}| \tag{5}$$

where $\boldsymbol{G}_{k,j}$ denotes the entry of the Gram matrix $\boldsymbol{G} = \boldsymbol{D}^{\mathrm{T}}\boldsymbol{D}$.

Zadeh *et al.* [29] show that a sparse decomposition with $\|\boldsymbol{\alpha}_0\|_0 < (1/2)\mathrm{spark}(\boldsymbol{D})$ is unique, but its stability against noise $\boldsymbol{n}$ has been proved only for highly more restrictive decompositions satisfying $\|\boldsymbol{\alpha}_0\|_0 < (1 + M^{-1})/2$, because usually $(1 + M^{-1})/2 \ll (1/2)\mathrm{spark}(\boldsymbol{D})$. The $\mathrm{spark}(\boldsymbol{D})$ here is the smallest number of columns from $\boldsymbol{D}$ that are linearly-dependent.

These works use worst-case reasoning exclusively, deriving conditions which must apply to every dictionary, every sparse representation and every bounded noise vector [28]. Therefore, such conditions, generally used for analysing the performance of sparse coding with a fixed dictionary, are unduly restrictive and excessive pessimism for many practical applications. In the next section, we will establish a SDL method via using similar but more relaxed strategy to learn the adaptive dictionary.

3 Radar HRRP target recognition based on SDL

In this section, we will explain the existent issues in the practical applications and then design a method, referred to as SDL, for HRRP-based RATR.

3.1 Characteristics of HRRP

As previously mentioned, to capture all potential data patterns, following the theory of sparse representation, the dictionary is routinely designed to be overcomplete thereby redundant, which makes the sparse representations to be unstable with inaccurate measurements.

For HRRP-based RATR, according to the scattering centre model [9, 10], the variation of the target aspect will cause different range shifts for different scattering centres on the target, even within the aspect region where the scattering centre structure remains unchanged. This phenomenon is target-aspect sensitivity, which leads to unknown amplitude perturbations among the HRRPs in a frame. Here an HRRP frame is a small target-aspect sector without scatterers' motion through range cells (MTRC). In addition, the amplitude fluctuation widely existed in the wideband detection, noise spikes and outliers will also make the measurements inaccurate.

The measured HRRPs and the correlation coefficient between the adjacent HRRPs in a frame for different targets are illustrated in Fig. 1. It is reasonable that HRRP can be composed of two components: (i) strong correlation component, which depends on

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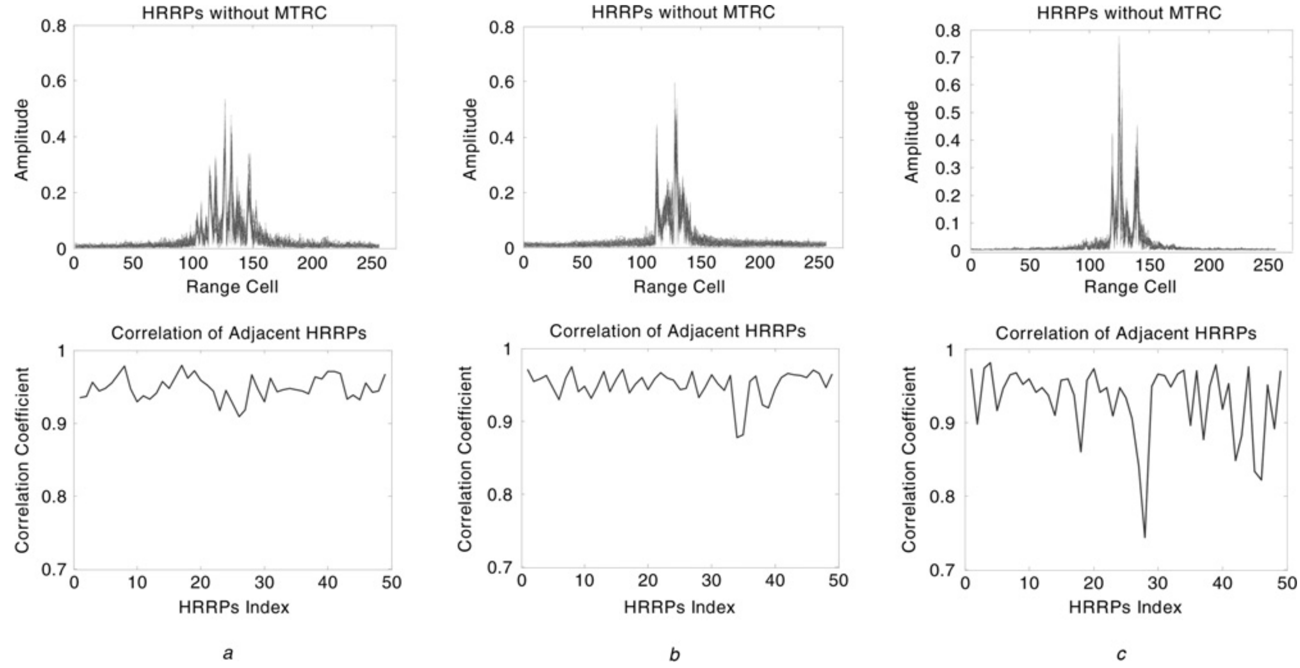

Fig. 1 *Measured HRRPs in a frame and the correlation coefficient between the adjacent HRRPs for different airplane targets*

The shades code the amplitude perturbations
a Yark-42
b Cessna Citation S/II
c An-26

target's scatterer centre structure, and is stable since the scattering centre structure remains approximately unchanged (see the first row of Fig. 1); (ii) unknown amplitude perturbation component, which mainly depends on the variation of aspect, and are unstable and similar to the 'coherent speckles' in SAR imaging. These amplitude perturbations may cause the uncertainty and ambiguity of the sparse representation, and ultimately affect the recognition performance when using dictionary learning methods.

3.2 Stability conditions in the presence of amplitude perturbations

For HRRP-based RATR, the sparse representations of adjacent HRRPs in a frame may be staggeringly different due to the amplitude perturbations. Expanding the conclusions in [28], the stability conditions in the presence of amplitude perturbations can be presented accordingly:

3.2.1 Ideal noiseless: With a given overcomplete dictionary $\boldsymbol{D} \in \mathbb{R}^{P\times K}$ and $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]$ denoting a set of ideal noiseless HRRPs in a frame, we make a simplified but reasonable assumption that the strong correlation component of each HRRP is approximately the same, that is, $\boldsymbol{x}_i = \boldsymbol{x}_0 + \boldsymbol{v}_i = \boldsymbol{D}\boldsymbol{\alpha}_0 + \boldsymbol{v}_i = \boldsymbol{D}\boldsymbol{\alpha}_i$, $\forall i \in \{1, 2, \ldots, N\}$, where $\boldsymbol{x}_0$ is the strong correlation component and $\boldsymbol{v}_i$ denotes the unknown amplitude perturbation in the ith HRRP sample $\boldsymbol{x}_i$. We assume that $\boldsymbol{\alpha}_0$, the sparse representation of $\boldsymbol{x}_0$, has at most L ($L \ll K$) non-zero elements.

If the condition $\|\boldsymbol{\alpha}_0\|_0 < (1 + M^{-1})/2$ can be satisfied, then $\boldsymbol{\alpha}_0$ is the unique sparsest representations of $\boldsymbol{x}_0$, and $\boldsymbol{\alpha}_i$ (the sparse representation of $\boldsymbol{x}_i$) obeys

$$\mathrm{supp}(\boldsymbol{\alpha}_i) = \mathrm{supp}(\boldsymbol{\alpha}_j), \quad \forall i \neq j \in \{0, 1, \ldots, N\} \tag{6}$$

$$r(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_0) = \|\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_0\|_2^2 \leq \varepsilon_i = \frac{4\|\boldsymbol{v}_i\|_2^2}{1 - M(2L-1)} \tag{7}$$

where $\mathrm{supp}(\boldsymbol{\alpha})$ denotes the position of non-zero elements in vector $\boldsymbol{\alpha}$.

3.2.2 Case with noise: Given a set of noiseless HRRPs $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]$, their noisy observations are $\boldsymbol{y}_i = \boldsymbol{x}_i + \boldsymbol{n}_i$, $\forall i \in \{1, 2, \ldots, N\}$. $\boldsymbol{\alpha}_i$ is the sparse representation of the noisy signal $\boldsymbol{y}_i$ by solving (2). We consider the model $\boldsymbol{y}_i = \boldsymbol{x}_0 + \boldsymbol{v}_i + \boldsymbol{n}_i = \boldsymbol{D}\boldsymbol{\alpha}_0 + \boldsymbol{v}_i + \boldsymbol{n}_i = \boldsymbol{D}\boldsymbol{\alpha}_i$, where $\boldsymbol{n}_i$ is the noise vector in the ith observation. When the model satisfies $\|\boldsymbol{\alpha}_0\|_2 = L < (1 + M^{-1})/2$, then $\boldsymbol{\alpha}_0$ is the unique sparsest such representation of $\boldsymbol{x}_0$, and the sparse representation $\boldsymbol{\alpha}_i$ obeys

$$\mathrm{supp}(\boldsymbol{\alpha}_i) = \mathrm{supp}(\boldsymbol{\alpha}_j), \quad \forall i \neq j \in \{0, 1, \ldots, N\} \tag{8}$$

$$r(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_0) = \|\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_0\|_2^2 \leq \varepsilon_i = \frac{4\|\boldsymbol{v}_i + \boldsymbol{n}_i\|_2^2}{1 - M(2L-1)} \tag{9}$$

These stability conditions are too restrictive to reach, since the mutual coherence M of the dictionary may be very large due to the strong correlation between the adjacent HRRPs. Moreover, these conclusions only deal with the case when the dictionary is fixed. Consequently, for adaptive dictionary, we will establish a SDL method using similar but more relaxed strategy in the next section.

3.3 Method construction

A stable method is designed, which is referred to as SDL, for radar HRRP target recognition. The basic idea is that the amplitude perturbations among the adjacent HRRPs are handled by enforcing the structured constraints that the sparse representations should have the same support and lower variance. SDL is consisted of two phases as follow:

3.3.1 Training phase: To model the HRRP dataset $\boldsymbol{X}_c \in \mathbb{R}^{P\times N_c}$ from target c ($c \in \{1, 2, \ldots, C\}$ with C denoting the number of targets), we divide the training dataset $\boldsymbol{X}_c$ into L_c frame, that is, $\boldsymbol{X}_c = [\boldsymbol{X}_{c,1}, \boldsymbol{X}_{c,2}, \ldots, \boldsymbol{X}_{c,L_c}] \in \mathbb{R}^{P\times N_c}$. Let $\boldsymbol{X}_{c,l} = \{\boldsymbol{x}_{c,l}^i\}_{i=1}^{N_{c,l}}$ be the HRRPs in the lth frame from target c, where $N_{c,l}$ denotes the number of samples in this frame. The columns of the initial class dictionary $\boldsymbol{D}_\mathrm{c} = [\boldsymbol{d}_1, \boldsymbol{d}_2, \ldots, \boldsymbol{d}_K] \in \mathbb{R}^{P\times K}$ are normalised by dividing their l_2-norm respectively to avoid degenerate solution.

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The mathematical model adapted in SDL is given by

$$\boldsymbol{x}_{c,l}^{i} = \boldsymbol{D}_{\mathrm{c}}\boldsymbol{\alpha}_{c,l}^{i} + \boldsymbol{\mu}_{c} + \boldsymbol{n}_{c,l}^{i} \tag{10}$$

where $\boldsymbol{\mu}_c = 1/N_{c,l}\sum_{i=1}^{N_{c,l}}\boldsymbol{x}_{c,l}^{i}$ denotes the mean vector of the training samples from target c, $\boldsymbol{n}_{c,l}^{i}$ is the error vector of the ith sample in the lth subset from target c.

Given a set of training samples, we seek the dictionary that leads to the best representation for each member in this set, under several constrains. Learning the class dictionaries $\boldsymbol{D}_{\mathrm{c}}^{*}$ with a fixed number K of atoms is addressed by solving the following minimisation problem.

$$\begin{aligned}\{\boldsymbol{D}_c, \boldsymbol{A}_c\} = \arg\min_{\boldsymbol{D}_c,\boldsymbol{A}_c} \sum_{l=1}^{L_c}&\left\{\left(\sum_{i=1}^{N_{c,l}}\left\|\boldsymbol{x}_{c,l}^{i} - \boldsymbol{\mu}_c - \boldsymbol{D}_c\boldsymbol{\alpha}_{c,l}^{i}\right\|_2^2\right)\right.\\ &\left.+\frac{\lambda}{N_{c,l}}\sum_{i=1}^{N_{c,l}}\left\|\boldsymbol{\alpha}_{c,l}^{i} - \boldsymbol{m}_{c,l}\right\|_2^2\right\}\\ \text{s.t.}\quad \left\|\boldsymbol{\alpha}_{c,l}^{i}\right\|_0 \le S_{tr},\quad &\left\|\boldsymbol{d}_c^{k}\right\|_2^2 = 1\\ &\mathrm{supp}(\boldsymbol{\alpha}_{c,l}^{i}) = \mathrm{supp}(\boldsymbol{\alpha}_{c,l}^{j})\\ &\boldsymbol{m}_{c,l} = \frac{1}{N_{c,l}}\sum_{i=1}^{N_{c,l}}\boldsymbol{\alpha}_{c,l}^{i}\\ &\forall l \in \{1, 2, \ldots, L_c\},\ i, j \in \{1, 2, \ldots, N_{c,l}\},\\ &k \in \{1, 2, \ldots, K\}\end{aligned} \tag{11}$$

where $\boldsymbol{A}_c = \{\boldsymbol{A}_{c,l}\}_{l=1}^{L_c}$ with $\boldsymbol{A}_{c,l} = [\boldsymbol{\alpha}_{c,l}^{1}, \boldsymbol{\alpha}_{c,l}^{2}, \ldots, \boldsymbol{\alpha}_{c,l}^{N_{c,l}}]$ is the sparse representations in the lth frame from target c, the non-negative parameter λ ($\lambda \ge 0$) controls the trade-off between the reconstruction error term (the first term of the objective function in (11)) and the variance term (the second term). S_{tr} represents the training sparsity level. The vector $\boldsymbol{m}_{c,l} \in \mathbb{R}^{K\times 1}$ is the mean vector of the sparse representations in the lth frame from target c.

In (11), different classes could have different non-zero locations in sparse representation matrix. In the same class, only the HRRPs in a small target-aspect sector without scatterers' MTRC ($\boldsymbol{X}_{c,l}$, we call it a frame) have similar target's scatterer centre structure, hence we constrain that the adjacent training samples in one frame with the same class have the same non-zero locations.

3.3.2 Classification phase: We denote $\boldsymbol{y} \in \mathbb{R}^{P\times 1}$ as a test sample, whose sparse representation $\boldsymbol{\beta}_c^{*} \in \mathbb{R}^{K\times 1}$ with respect to the learned dictionary $\boldsymbol{D}_{\mathrm{c}}^{*}$ can be solved by

$$\boldsymbol{\beta}_c^{*} = \arg\min_{\boldsymbol{\beta}_c}\left\|\boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}_{\mathrm{c}}^{*}\boldsymbol{\beta}_c\right\|_2 \ \text{s.t.}\ \left\|\boldsymbol{\beta}_c\right\|_0 \le S_{\mathrm{te}} \tag{12}$$

S_{te} represents the test sparsity level. However, as mentioned above, (12) is sensitive to the amplitude perturbations, thus we employ a more robust loss function, which enforces the structured regularisation to select the optimal compacted dictionary basis vectors, to learn $\boldsymbol{\beta}_{c,l}^{*}$.

$$\boldsymbol{\beta}_{c,l}^{*} = \arg\min_{\boldsymbol{\beta}_{c,l}}\left\|\boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}_{c,l}^{*}\boldsymbol{\beta}_{c,l}\right\|_2 \tag{13}$$

Upon solution, let $D_{\mathrm{c},l}^{*} = \{\boldsymbol{d}_c^{k} \mid \|\boldsymbol{\alpha}_{c,l}^{k:}\|_2 > 0\}$, where $\boldsymbol{\alpha}_{c,l}^{k:}$ is the kth row of the learned sparse representation matrix of the training data $\boldsymbol{X}_c$ and $\boldsymbol{d}_c^{k}$ is the kth column of $\boldsymbol{D}_{\mathrm{c}}^{*}$. We design the class sub-dictionary $\boldsymbol{D}_{\mathrm{c},l}^{*} \in \mathbb{R}^{P\times|D_{\mathrm{c},l}^{*}|}$ in the lth frame from target c using all $\boldsymbol{d}_c^{k} \in D_{\mathrm{c},l}^{*}$ as its atoms. Obviously, (13) is a least square problem that has a closed-form solution. To combine the virtues of 'sparsity', we can construct the loss function by a sparsity-inducing term.

$$\boldsymbol{\beta}_{c,l}^{*} = \arg\min_{\boldsymbol{\beta}_{c,l}}\left\|\boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}_{\mathrm{c},l}^{*}\boldsymbol{\beta}_{c,l}\right\|_2 \ \text{s.t.}\ \left\|\boldsymbol{\beta}_{c,l}\right\|_0 \le S_{\mathrm{te}} \tag{14}$$

The optimal frame index $\hat{l}_c$ for target c is

$$\hat{l}_c = \arg\min_{l=1,2,\ldots,L_c}\left\|\boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}_{\mathrm{c},l}^{*}\boldsymbol{\beta}_{c,l}^{*}\right\|_2 \tag{15}$$

When the sparse representation $\boldsymbol{\beta}_{c,l}^{*}$ with respect to the class sub-dictionary $\boldsymbol{D}_{\mathrm{c},l}^{*}$ are learned, based on the minimum squared reconstruction error approximation, the optimal class label of $\boldsymbol{y}$ can be estimated as

$$c^{*} = \arg\min_{c=1,2,\ldots,C}\left\|\boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}_{\mathrm{c},\hat{l}_c}^{*}\boldsymbol{\beta}_{c,\hat{l}_c}^{*}\right\|_2 \tag{16}$$

4 Training and classification algorithms for SDL

Because the objective function in (11) comprises multiple complex constrains, it is different to solve in general by existing optimisation algorithms. Hence we derive several efficient algorithms for the implementation of SDL.

4.1 Training algorithm for SDL

For the non-convex optimisation problem in (11), we make use mainly of GA for sparse coding because of its simplicity and efficiency. Next, a least-squares problem with l_2-norm constraint is proposed for dictionary update.

4.1.1 First stage: stable sparse coding: $\boldsymbol{D}_{\mathrm{c}}$ is the cth class dictionary with l_2 normalised columns and $\boldsymbol{X}_{c,l} = \{\boldsymbol{x}_{c,l}^{i}\}_{i=1}^{N_{c,l}}$ denotes the training dataset in the lth frame from target c. Instead of addressing all training data together, we get a decoupling of the minimisation task to many smaller ones, each of the form handling one HRRP frame.

$$\begin{aligned}\boldsymbol{A}_{c,l} = \arg\min_{\boldsymbol{A}_{c,l}}&\sum_{i=1}^{N_{c,l}}\left\|\boldsymbol{x}_{c,l}^{i} - \boldsymbol{\mu}_c - \boldsymbol{D}_c\boldsymbol{\alpha}_{c,l}^{i}\right\|_2^2 + \frac{\lambda}{N_{c,l}}\sum_{i=1}^{N_{c,l}}\left\|\boldsymbol{\alpha}_{c,l}^{i} - \boldsymbol{m}_{c,l}\right\|_2^2\\ \text{s.t.}\quad &\left\|\boldsymbol{\alpha}_{c,l}^{i}\right\|_0 \le S_{\mathrm{tr}}\\ &\mathrm{supp}(\boldsymbol{\alpha}_{c,l}^{i}) = \mathrm{supp}(\boldsymbol{\alpha}_{c,l}^{j})\\ &\boldsymbol{m}_{c,l} = \frac{1}{N_{c,l}}\sum_{i=1}^{N_{c,l}}\boldsymbol{\alpha}_{c,l}^{i}\\ &\forall i, j \in \{1, 2, \ldots, N_{c,l}\}\end{aligned} \tag{17}$$

Based on the greedy strategy [23], we state the optimisation problem (17) as

$$\begin{aligned}\min_{\Lambda_{c,l},|\Lambda_{c,l}|\le T}&\sum_{i=1}^{N_{c,l}}\left\|\boldsymbol{x}_{c,l}^{i} - \boldsymbol{\mu}_c - \boldsymbol{D}_{\Lambda_{c,l}}\left[\boldsymbol{D}_{\Lambda_{c,l}}^{\mathrm{T}}\boldsymbol{D}_{\Lambda_{c,l}}\right]^{-1}\boldsymbol{D}_{\Lambda_{c,l}}^{\mathrm{T}}\boldsymbol{x}_{c,l}^{i}\right\|_2^2\\ &+\frac{\lambda}{N_{c,l}}\sum_{i=1}^{N_{c,l}}\left\|\left[\boldsymbol{D}_{\Lambda_{c,l}}^{\mathrm{T}}\boldsymbol{D}_{\Lambda_{c,l}}\right]^{-1}\boldsymbol{D}_{\Lambda_{c,l}}^{\mathrm{T}}\boldsymbol{x}_{c,l}^{i} - \boldsymbol{m}_{c,l}\right\|_2^2\end{aligned} \tag{18}$$

where indices $\Lambda_{c,l} = \mathrm{supp}(\boldsymbol{A}_{c,l})$.

Considering that the sparse representation is $\boldsymbol{\alpha}_{c,l}^{i} = [\boldsymbol{D}_{\Lambda_{c,l}}^{\mathrm{T}}\boldsymbol{D}_{\Lambda_{c,l}}]^{-1}\boldsymbol{D}_{\Lambda_{c,l}}^{\mathrm{T}}\boldsymbol{x}_{c,l}^{i}$. A GA is proposed for this optimisation problem, which is summarised in Algorithm 1:

Algorithm 1: Stable sparse coding (SSC)

Input: The cth class dictionary $\boldsymbol{D}_{\mathrm{c}} \in \mathbb{R}^{P\times K}$, training dataset in the lth frame from target c $\boldsymbol{X}_{c,l} = \{\boldsymbol{x}_{c,l}^{i}\}_{i=1}^{N_{c,l}}$, training sparsity level S_{tr};
Convention: Λ is an empty set, $\boldsymbol{D}_{\Lambda}$ is an empty matrix,

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$n = 0$;
Output: The stable sparse representation matrix $\boldsymbol{A}_{c,l}$ and indices $\Lambda_{c,l}$.
Algorithm steps:

1. Initialise the residual of the ith training sample $\boldsymbol{r}^i = \boldsymbol{x}^i_{c,l} - \boldsymbol{\mu}_c$;
2. Find the index β satisfying

$$\beta = \arg \max_{k=1,2,\ldots,K} f(\boldsymbol{d}^k_c) = \sum_{i=1}^{N_{c,l}} \left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i \right)^2 - \frac{\lambda}{N_{c,l}} \sum_{i=1}^{N_{c,l}} \left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i - m_k \right)^2$$

3. Update sets $\Lambda := \Lambda \cup \{\beta\}$, $\boldsymbol{D}_\Lambda := [\boldsymbol{D}_\Lambda, \boldsymbol{d}^\beta_c]$;
4. Compute new sparse representations, reconstructions, and residuals by

$$\boldsymbol{\alpha}^i_{c,l} = \arg \min_{\boldsymbol{\alpha}} \left\| \boldsymbol{x}^i_{c,l} - \boldsymbol{\mu}_c - \boldsymbol{D}_\Lambda \boldsymbol{\alpha} \right\|_2^2 = (\boldsymbol{D}^{\mathrm{T}}_\Lambda \boldsymbol{D}_\Lambda)^{-1} \boldsymbol{D}^{\mathrm{T}}_\Lambda (\boldsymbol{x}^i_{c,l} - \boldsymbol{\mu}_c)$$

$$\hat{\boldsymbol{x}}^i_{c,l} = \boldsymbol{D}_\Lambda \boldsymbol{\alpha}^i_{c,l} + \boldsymbol{\mu}_c, \quad \boldsymbol{r}^i = \boldsymbol{x}^i_{c,l} - \hat{\boldsymbol{x}}^i_{c,l} := \boldsymbol{r}^i - \boldsymbol{D}_\Lambda \boldsymbol{\alpha}^i_{c,l}$$

5. $n := n + 1$. If $n < S_{\mathrm{tr}}$, then go back to Step. 2;
6. Return estimates $\hat{\boldsymbol{x}}^i_{c,l}$, the stable sparse representation matrix $\boldsymbol{A}_{c,l} = [\boldsymbol{\alpha}^1_{c,l}, \boldsymbol{\alpha}^2_{c,l}, \ldots, \boldsymbol{\alpha}^{N_{c,l}}_{c,l}]$, indices $\Lambda_{c,\ l} = \Lambda$.

Step 2 of the Algorithm 1 is referred to as the greedy selection. The intuition behind maximising the value of function $f(\boldsymbol{d}^k_c)$ $(\forall k = \{1, 2, \ldots, K\})$ is that we wish to find an atom $\boldsymbol{d}_\beta$ that can minimise the value of objective function (17) stepwise, see (19).

$$\begin{aligned}
\beta &= \arg \min_{k=1,2,\ldots,K} \sum_{i=1}^{N_{c,l}} \left\| \boldsymbol{r}^i - \boldsymbol{d}^k_c \left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{d}^k_c \right)^{-1} (\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i \right\|_2^2 \\
&\quad + \frac{\lambda}{N_{c,l}} \sum_{i=1}^{N_{c,l}} \left(\left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{d}^k_c \right)^{-1} (\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i - m_k \right)^2 \\
&= \arg \min_{k=1,2,\ldots,K} \sum_{i=1}^{N_{c,l}} \left\| \boldsymbol{r}^i - \boldsymbol{d}^k_c (\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i \right\|_2^2 + \frac{\lambda}{N_{c,l}} \sum_{i=1}^{N_{c,l}} \left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i - m_k \right)^2 \\
&= \arg \max_{k=1,2,\ldots,K} \sum_{i=1}^{N_{c,l}} \left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i \right)^2 - \frac{\lambda}{N_{c,l}} \sum_{i=1}^{N_{c,l}} \left((\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{r}^i - m_k \right)^2
\end{aligned}$$

where $\boldsymbol{d}^{\mathrm{T}}_k \boldsymbol{d}_k = 1$ and m_k is the kth element of vector

$$\boldsymbol{m} = \frac{1}{N_{c,l}} \left[\sum_{i=1}^{N_{c,l}} \boldsymbol{d}^{\mathrm{T}}_1 \boldsymbol{r}^i, \ \sum_{i=1}^{N_{c,l}} \boldsymbol{d}^{\mathrm{T}}_2 \boldsymbol{r}^i, \ \ldots, \ \sum_{i=1}^{N_{c,l}} \boldsymbol{d}^{\mathrm{T}}_K \boldsymbol{r}^i \right]^{\mathrm{T}} \text{and}$$

$$(\boldsymbol{d}^k_c)^{\mathrm{T}} \boldsymbol{d}^k_c = 1.$$

4.1.2 Second stage: learning adaptive dictionary: Given the learned stable sparse representation matrix $\boldsymbol{A}_{c,l} = \{\boldsymbol{\alpha}^i_{c,l}\}_{i=1}^{N_{c,l}}$ with $l \in \{1, 2, \ldots, L_c\}$, the cth class dictionary $\boldsymbol{D}_{\mathrm{c}}$ can be updated by

$$\begin{aligned}
\boldsymbol{D}_{\mathrm{c}} &= \arg \min_{\boldsymbol{D}_{\mathrm{c}}} \ \sum_{l=1}^{L_c} \sum_{i=1}^{N_{c,l}} \left\| \boldsymbol{x}^i_{c,l} - \boldsymbol{\mu}_c - \boldsymbol{D}_{\mathrm{c}} \boldsymbol{\alpha}^i_{c,l} \right\|_2^2 \\
&\text{s.t.} \quad \forall k, \quad \left\| \boldsymbol{d}_k \right\|_2^2 = 1
\end{aligned} \tag{20}$$

Let $\boldsymbol{X}_c = [\boldsymbol{X}_{c,1}, \ldots, \boldsymbol{X}_{c,L_c}]$, $\boldsymbol{A}_c = [\boldsymbol{A}_{c,1}, \ldots, \boldsymbol{A}_{c,L_c}]$ and $\boldsymbol{U}_c = [\boldsymbol{\mu}_c, \ldots, \boldsymbol{\mu}_c] \in \mathbb{R}^{P \times N_c}$, we can rewrite (20) in a more succinct form using matrices as following

$$\begin{aligned}
\boldsymbol{D}_{\mathrm{c}} &= \arg \min_{\boldsymbol{D}_c} \ \left\| \mathbf{X}_c - \boldsymbol{U}_c - \boldsymbol{D}_{\mathrm{c}} \boldsymbol{A}_c \right\|_{\mathrm{F}}^2 \\
&\text{s.t.} \quad \forall k, \quad \left\| \boldsymbol{d}^k_c \right\|_2^2 = 1
\end{aligned} \tag{21}$$

A least-squares problem with l_2-norm constraint is solved for updating all atoms of dictionaries. The solution of the optimisation problem (21) can be expressed as

$$\boldsymbol{D}^*_{\mathrm{c}} = \mathrm{norm}\left((\boldsymbol{X}_c - \boldsymbol{U}_c) \boldsymbol{A}^{\mathrm{T}}_c \left(\boldsymbol{A}_c \boldsymbol{A}^{\mathrm{T}}_c \right)^{-1} \right) \tag{22}$$

where norm$(\cdot)$projects $(\boldsymbol{X}_c - \boldsymbol{U}_c)\boldsymbol{A}^{\mathrm{T}}_c(\boldsymbol{A}_c\boldsymbol{A}^{\mathrm{T}}_c)^{-1}$ to a matrix with l_2 normalised columns.

In Algorithm 1, we first divide the training dataset $\boldsymbol{X}_c$ from target c into K continuous data segments, that is, $\boldsymbol{X}_c = [\boldsymbol{X}_{c,1}, \boldsymbol{X}_{c,2}, \ldots, \boldsymbol{X}_{c,K}]$. The initialisation of the atoms in the class dictionary $\boldsymbol{D}_{\mathrm{c}}$ can be calculated from the average of the adjacent training samples.

$$\boldsymbol{d}^k_c = \frac{1}{|\boldsymbol{X}_{c,k}|} \sum_{i=1}^{|\boldsymbol{X}_{c,k}|} \boldsymbol{x}^i_{c,k} - \boldsymbol{\mu}_c, \quad \forall k = 1, \ldots, K \tag{23}$$

where $|\boldsymbol{X}_{c,k}|$ is the number of training samples in $\boldsymbol{X}_{c,k}$. In later iterations, we utilise the updated dictionary $\boldsymbol{D}_{\mathrm{c}}$ calculated by (21) as the input of the Algorithm 1.

4.2 Classification algorithm for SDL

In classification phase, a test sample $\boldsymbol{y}$ is submitted to all learned class dictionaries. Using the learned class dictionary $\boldsymbol{D}^*_{\mathrm{c}}$, the classical sparse coding algorithm for calculating sparse representation is described as (12), which may be sensitive to the amplitude perturbations. To deal with this issue, we develop two robust loss functions (see (13) and (14)) via enforcing the structured regularisation to select the optimal sub-dictionary basis vectors. The corresponding procedure for classification phase is summarised in Algorithm 2:

Algorithm 2: Classification algorithm for SDL

Input: A test sample $\boldsymbol{y} \in \mathbb{R}^{P \times 1}$, the learned class dictionaries $\boldsymbol{D}^*_{\mathrm{c}}$ with $c \in \{1, 2, \ldots, C\}$, sparse representation matrix $\boldsymbol{A}_{c,l}$ of training data, test sparsity level S_{te};
Convention: Normalise y to have unit l_2-norm.
Output: the class label $I(\boldsymbol{y})$.

Algorithm steps:

1. Let $D^*_{\mathrm{c},l} = \{\boldsymbol{d}^k_c | \ \|\boldsymbol{\alpha}^{k:}_{c,l}\|_2 > 0\}$, where $\boldsymbol{\alpha}^{k:}_{c,l}$ is the kth row of $\boldsymbol{A}_{c,l}$. The lth frame from the cth class sub-dictionary $\boldsymbol{D}^*_{\mathrm{c},l} \in \mathbb{R}^{P \times |D^*_{\mathrm{c},l}|}$ is designed via using all $\boldsymbol{d}^k_c \in D^*_{\mathrm{c},l}$ as its atoms.
2. Solve the minimum squared problem

$$\boldsymbol{\beta}^*_{c,l} = \arg \ \min_{\boldsymbol{\beta}_{c,l}} \left\| \boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}^*_{\mathrm{c},l} \boldsymbol{\beta}_{c,l} \right\|_2$$

or, solve the l_0-minimisation problem

$$\boldsymbol{\beta}^*_{c,l} = \arg \ \min_{\boldsymbol{\beta}_{c,l}} \left\| \boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}^*_{\mathrm{c},l} \boldsymbol{\beta}_{c,l} \right\|_2 \text{s.t.} \left\| \boldsymbol{\beta}_{c,l} \right\|_0 \le S_{\mathrm{te}}$$

3. Compute the optimal frame index for target c

$$\hat{l}_c = \arg \min_{l=1,2,\ldots,L_c} \left\| \boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}^*_{\mathrm{c},l} \boldsymbol{\beta}^*_{c,l} \right\|_2$$

Output:

$$I(\boldsymbol{y}) = \arg \min_{c=1,2,\ldots,C} \left\| \boldsymbol{y} - \boldsymbol{\mu}_c - \boldsymbol{D}^*_{\mathrm{c},\hat{l}_c} \boldsymbol{\beta}^*_{c,\hat{l}_c} \right\|_2.$$

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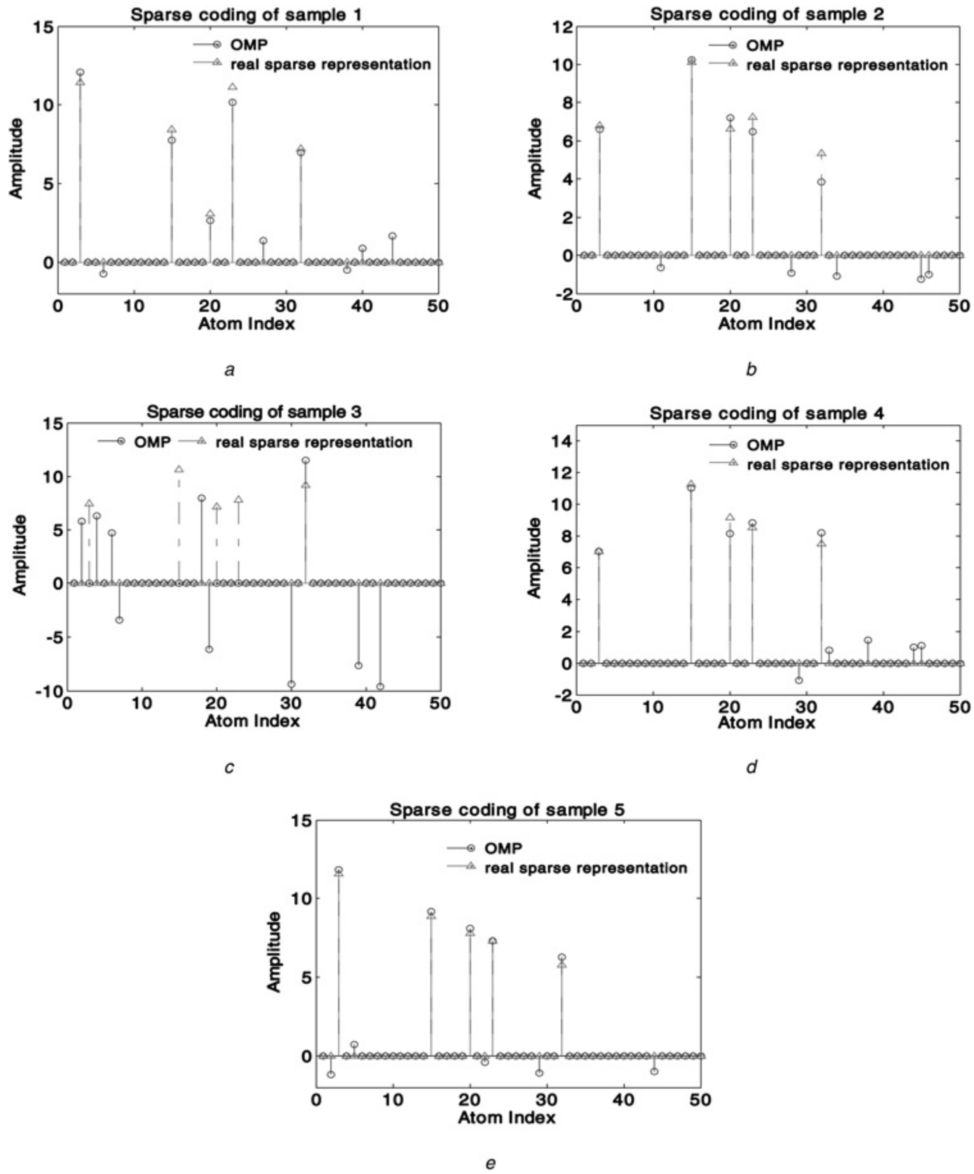

Fig. 2 *Synthetic result: real sparse representations of signals and the sparse representations via OMP in the presence of amplitude perturbations*

5 Experimental results

In this part, we will examine the performance of the proposed method on the following synthetic data and measured HRRP dataset.

5.1 Results on synthetic data

Each column in the random fixed dictionary $\boldsymbol{D}$ of size 20×50 is generated with uniform spherical ensemble and normalised to a unit l_2-norm. We use a randomly-generated matrix $\boldsymbol{A} = \{\boldsymbol{\alpha}_i\}_{i=1}^{N}$ satisfying $\|\boldsymbol{\alpha}_i\|_0 = 5$ and $\text{supp}(\boldsymbol{\alpha}_i) = \text{supp}(\boldsymbol{\alpha}_j),\ \forall i, j \in \{1, 2, \ldots, N\}$ as the ideal sparse representations from one class. The non-zero entries of $\boldsymbol{A}$ are located in uniform random positions, and the values of those entries are drawn from a normal distribution. Then zero-mean white Gaussian noise with 15 dB signal-to-noise ratio (SNR) is added to the resulted signals as stochastic amplitude perturbations, obtaining synthetic data $\boldsymbol{x}_i = \boldsymbol{D}\boldsymbol{\alpha}_i + \boldsymbol{v}_i$. In this experiment, we choose $N = 5$, $\lambda = 1.5$ and try the SSC (Algorithm 1) on synthetic data to test the stability property in the presence of amplitude perturbations. Since the sparsity of signals is unknown in practice, we implement all algorithms with a sparsity level $S = 10$ to avoid the loss of information in the signals, which is slightly larger than the real number of non-zero entries.

In Fig. 2, OMP does not find accurate locations of the atoms and generates 'false' atoms in case 3. Hence OMP cannot effectively recover all signals in the presence of stochastic amplitude perturbations. Simultaneous orthogonal matching pursuit (SOMP) [23] is a common used group sparse coding algorithm. Fig. 3*a* presents the sparse representations by SOMP. Compared with OMP, SOMP reduces the number of 'false' atoms, but it also cannot accurately represent signals in the presence of amplitude perturbations. In Fig. 3*b*, the comparison between SOMP and SSC demonstrates that our algorithm cannot only find the accurate atoms effectively, but also recover signals accurately. We also analyse the performance of all algorithms with different sparsity levels ($S \geq 5$). While with increasing S, the recovered number and locations of the real sparse representations are nearly unchanged and the additional atoms just describe the amplitude perturbations in data. Thus, without loss of generality, we only present the results with a fixed sparsity levels ($S = 10$) in the experiment.

In sparse coding, a 'false' sparse representation may still guarantee a small reconstruction error of the input through a linear operation with an overcomplete dictionary $\boldsymbol{D}$. However, for recognition tasks, only the underlying sparse representations are useful for learning the appropriate dictionary. Thus, the recognition

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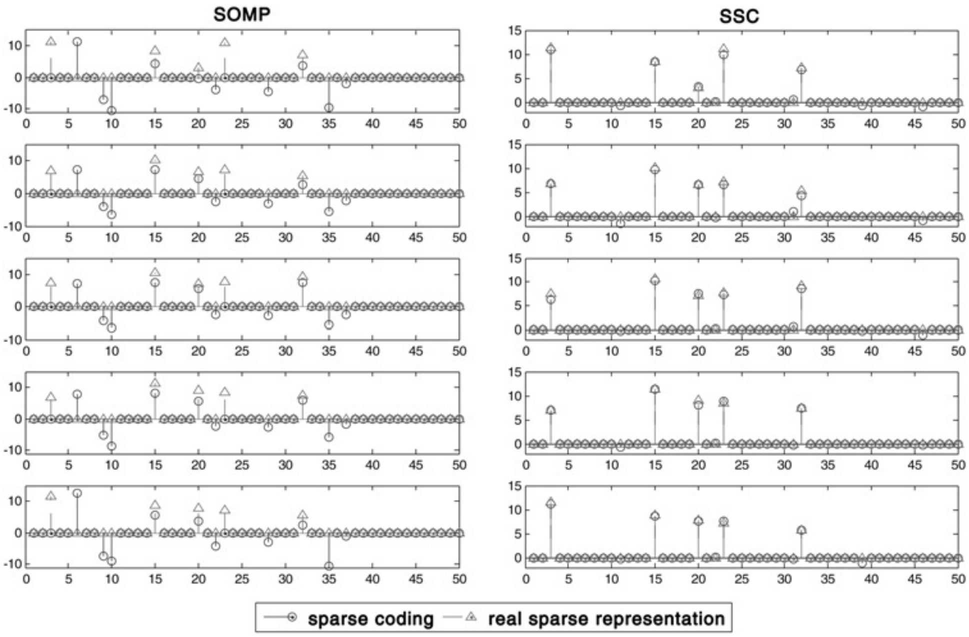

Fig. 3 *Left-hand side: the sparse representations of signals via SOMP in the presence of amplitude perturbations. Right-hand side: the sparse representations of signals via SSC in the presence of amplitude perturbations*

The first row is set for sample 1, the next for sample 2, and so on

performance of algorithms depends on the ability to find the underlying stable sparse representations.

This experiment confirms our intuition that given several observations with the same support in underlying sparse representations, we can improve the ability to identify the stable sparse representations, even in the presence of stochastic amplitude perturbations.

5.2 Results on measured HRRP dataset

5.2.1 Data description: We examine the performance of SDL on the measured HRRP data from three real airplanes, which are extensively used in [1, 3–5, 8–13]. The parameters of the airplane targets and measurement radar are shown in Table 1 and the projections of target trajectories onto the ground plane are shown in Fig. 4.

We take the data from the 2nd segment and a part of the 5th segment of Yark-42, the 6th and the 7th segments of Cessna Citation S/II, the 5th and the 6th segments of An-26 as the training samples. Other data segments are taken as test samples in the experiments. The measured HRRP is a 256-dimensional vector with the number of targets $C = 3$.

Since this HRRP dataset is measured under high SNR conditions, we assume that the training data are under high SNR and the test samples are got under the different condition of SNR as discussed in [12]. To evaluate the noise-robustness of the proposed method, we add simulated white noises to the inphase and quadrature components of the raw complex HRRP test data. The SNR is defined as

$$\mathrm{SNR} = 10 \times \lg\left(\frac{\bar{P}_s}{P_\mathrm{n}}\right) = 10 \times \lg\left(\frac{\sum_{i=1}^{I} P_{si}}{I \times P_\mathrm{n}}\right) \tag{24}$$

where $\bar{P}_s$ denotes the average power of original HRRP, P_{si} is the power of the original echo per range cell, I denotes the number of range cells, and P_n denotes the power of noise.

Table 1 Parameters of planes and radar in the ISAR experiment

Radar parameters	Centre frequency	5520 MHz	
	Bandwidth	400 MHz	
Planes	Length, m	Width, m	Height, m
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

5.2.2 Data preprocessing: A challenging task is how to deal with the time-shift and amplitude-scale sensitivity problems of HRRP [8, 12, 13]. We use the power spectra feature extracted from real HRRP data to eliminate time-shift sensitivity. Thus the dimension of the feature vector $P = 128$. To deal with this sensitivity, each power spectrum feature vector extracted from the real HRRP is normalised by dividing its l_2-norm.

5.2.3 Recognition performance: First, we utilise OMP as the fixed classification algorithm to examine the recognition performance of different dictionary learning methods used in training phase. Unless stated, there are 2800 samples for training, and the number of test samples is 1200 in the experiment. To deal with the imbalanced data set, the recognition accuracy can be calculated by

$$\text{Average recognition accuracy} = \frac{1}{C}\sum_{c=1}^{C}\frac{n_c}{N_c} \times 100\% \tag{25}$$

Here n_c is the number of test samples which are properly classified into the cth class. N_c is the number of test samples in the cth class.

A fixed classification algorithm, that is, OMP with sparsity level $S_\mathrm{te} = 3$, is utilised in the experiment to validate the effectiveness of the training algorithm of SDL. Compared with other traditional training methods in Table 2, our algorithm yields the best recognition performance (92.30%) because that SDL not only considers the structural similarity among adjacent HRRPs, but also is stable to against the amplitude perturbations existed in the adjacent HRRPs. K-SVD and MOD do not consider the structured information. Although the structural similarity is considered in SOMP, it is also unsuitable for solving stable sparse representation

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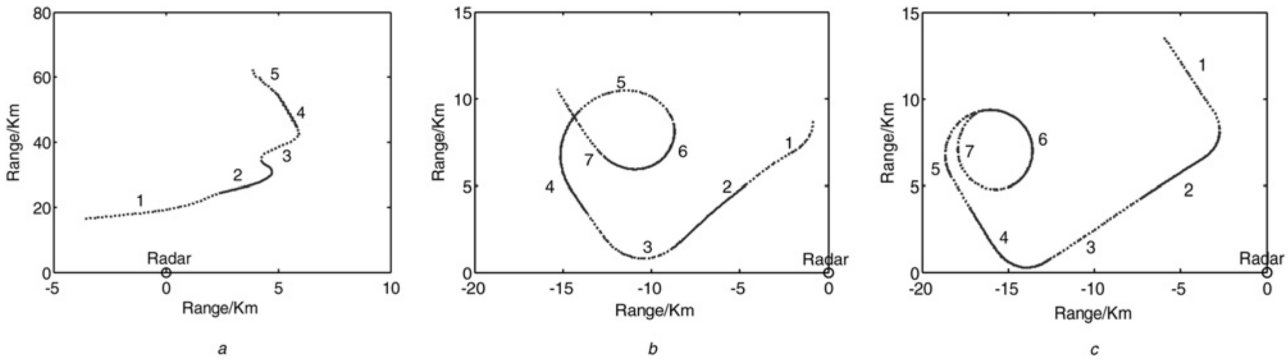

Fig. 4 *Projections of target trajectories onto the ground plane*
a Yark-42
b Cessna Citation S/II
c An-26

problem with inaccurate measurements. Note that, our algorithm is just a limiting special case of SDL, which uses OMP as its classification algorithm. We set the size of each class dictionary to be $P \times K$ where $P = 128$ and $K = 200$. The training sparsity level is $S_{tr} = 9$ for all methods. The parameters $N_{c,l}$ and λ ($N_{c,l} = 3$, $\lambda = 2.5$) in our algorithm are chosen via cross-validation on the training dataset. Thus the number of frames for Yark-42 is 234; the number of frames for Cessna Citation S/II and An-26 are both 350. Furthermore, our algorithm would reduce to SOMP when we set $\lambda = 0$.

Table 2 Recognition accuracies of different dictionary learning methods: K-SVD, MOD, SOMP and the training algorithm of SDL (%). The last one is a limiting case of SDL. All the methods use the same classification algorithm, that is, OMP

Train	Yark-42	CC S/II	An-26	Aver.
K-SVD	96.56	82.11	79.33	86.00
MOD	96.67	82.00	86.44	88.37
SOMP	95.33	87.56	88.67	90.52
SDL train	98.00	93.11	85.78	***92.30***

Table 3 Recognition accuracies of different classification algorithms: OMP and two proposed algorithms (%). The last two are the limiting cases of SDL with different classification algorithms. All the methods use the same training algorithm

Train + classif.	Yark-42	CC S/II	An-26	Aver.
SDL train + (12)	98.00	93.11	85.78	92.30
SDL train + (13)	97.67	97.33	81.56	**92.19**
SDL train + (14)	99.00	96.22	84.22	***93.15***

Table 3 presents the recognition accuracy of different classification algorithms mentioned in Section 3.3. Compared with OMP, the classification algorithms based on (13) and (14) impose structured information for learning representation of the test sample with respect to a compact sub-dictionary, which effectively avoid the instability caused by amplitude perturbations. The difference between (13) and (14) is that (13) is a least square problem, while there is a sparsity constraint in (14). In Table 3, the classification algorithm based on (14) has the best recognition accuracy, which validates that using structured regularisation to select the optimal sub-dictionary in classification phase can be helpful to improve the recognition performance. Moreover, it is worth noting that the recognition accuracy of the algorithm based on (13) does not outperform OMP because only an appropriate sparsity constraint may improve the recognition performance.

In the experiments below, we define SDL as the integration of the training algorithm mentioned in Section 4.1 and the classification algorithm based on (14). Since SDL uses a reconstruction strategy for recognition, we compare SDL with three traditional radar HRRP target recognition methods based on minimum reconstruction error approximation, that is, PCA-based minimum reconstruction error approximation [10], K-SVD [13] and MOD [14]. Fig. 5 shows that SDL has a better performance with a relative recognition accuracy improvement between 5 and 10%. Moreover, PCA has the best recognition accuracy (86.04%) in the methods using low-dimensional representations, which is slightly better than that of K-SVD (86%). This means that conventional dictionary learning methods may be affected by perturbations, thus leading to instability of sparse representations and limiting the recognition performance. Some parametric statistical models, that is, AGC [3] and FA [11], are also compared with further validate the effectiveness of our method.

Traditional methods usually work well when sufficient training data are available. The lack of the training data is a challenge for

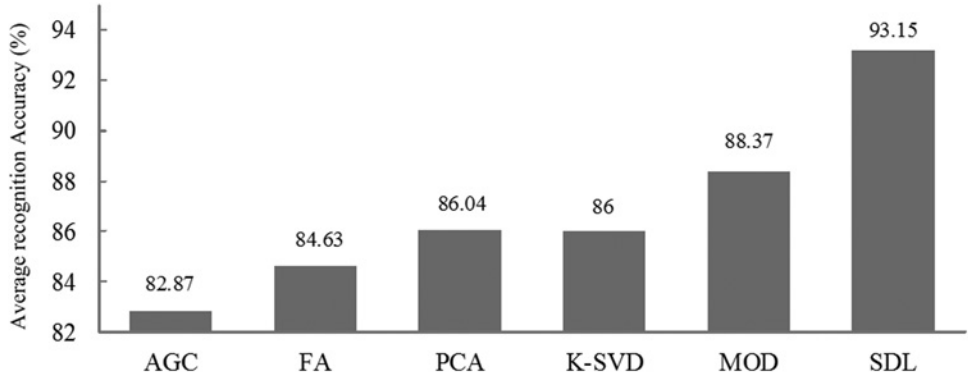

Fig. 5 *Average recognition accuracies of AGC, FA, PCA, K-SVD, MOD and SDL (%)*

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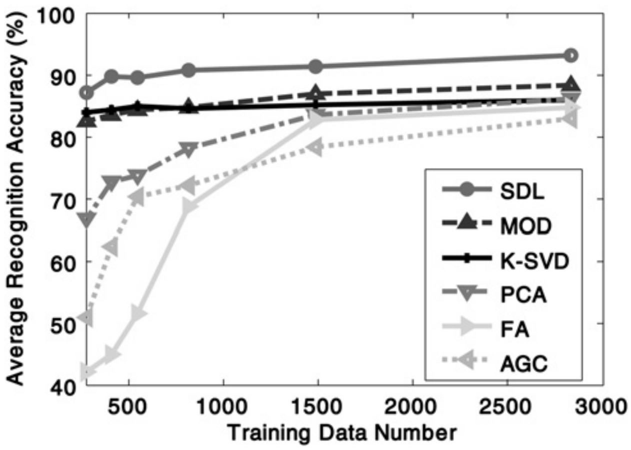

Fig. 6 *Variation of the recognition performance with training data size, via SDL, MOD, K-SVD, PCA, FA and AGC*

HRRP-based RATR. Especially, in the non-cooperative circumstance, for example, in the battlefield, the amount of data under test usually is huge, but the training data is limited. Such cases will lead to overfitting and restrict the generalisation performance of the recognition method. In the next experiment, we compare the recognition accuracy via training different methods with different training data sizes to validate their generalisation performance.

The recognition accuracies versus the size of training dataset are illustrated in Fig. 6. Note that, the HRRP dataset is measured under high SNR conditions (almost noiseless). According to the literatures [3, 10, 11, 13, 14], we do not add any noises to HRRPs in Fig. 6, that is, SNR $\simeq \infty$. Compared with other five methods, SDL notably improves the recognition performance even with very small training data size, which ensures a good generalisation capability. For SDL, K-SVD and MOD, when the training data size is very small ($N \leq 1500$), the recognition performance remains stable because the dictionary learning mechanism makes it possible to appropriately share the information among samples from different target-aspects and learn the global dictionary collectively; whereas for FA and AGC such training data size is not enough for the full statistical parameter estimation for the 128-length HRRP feature vector via separately learning from each target-aspect. Although PCA does not need to estimate parameters from each target-aspect, its low-dimensional representation space limits the recognition performance. SDL retains the structural similarity in representation space and overcome the uncertainty of sparse overcomplete representation in the presence of amplitude perturbations. Thus better recognition and generalisation performance with small data size is an advantage of SDL over other methods.

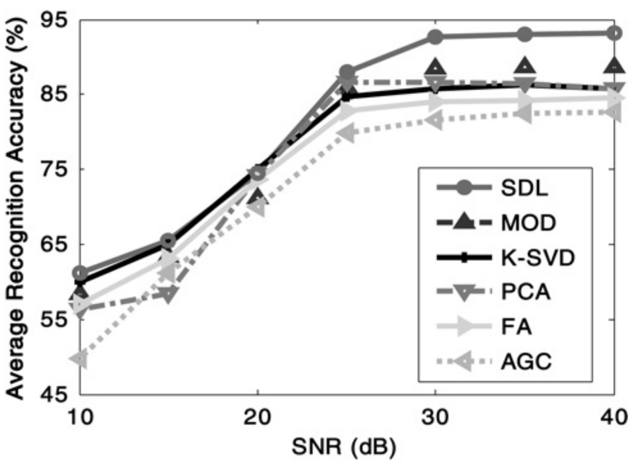

Fig. 7 *Variation of the recognition performance with SNR, SDL, MOD, K-SVD, PCA, FA and AGC*

Besides the amplitude perturbations, noises abound in the HRRP data by nature, and this requires more robustness on the learning method. Fig. 7 depicts the average recognition accuracy versus SNR, generated via the six HRRP recognition methods mentioned above. From Fig. 7, there is performance degradation to some extent for all the methods because the increase of noise leads the difference among the samples in different class decreases. When SNR $\geq$ 20 dB, SDL yields the best recognition accuracy. When SNR<20 dB, SDL's performance is similar to that of K-SVD, but is much better than that of other methods. This experiment confirms that our method is also robust against noise for measured HRRP data.

6 Conclusion

For radar HRRP target recognition task, conventional dictionary learning methods may cause uncertainty and ambiguity due to the redundancy of new space defined by overcomplete dictionary. In this paper, we develop SDL via utilising several stable strategies to learn adaptive dictionary. The amplitude perturbations in the proposed method are handled by enforcing a set of constraints that requires the sparse representations of adjacent HRRPs should have the same support and lower variance. The structural similarity in HRRPs is considered, thus improving the ability of finding the underlying sparse representations of HRRPs. Exploiting learned adaptive sub-dictionaries makes our method more stable and well suited to handle radar HRRP dataset. Experiments based on synthetic data and measured HRRP data show that SDL has an inspiring recognition and generalisation performance even with small training data size.

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